

**OPTIMISATION OF NANOSCALE CONVENTIONAL BULK
MOSFET USING TCAD
(CONTINUOUS ASSESSMENT AND MINI PROJECT**

by

JAISHREE V 24BEC1407

NAGULAPALLI MOUNICA SRINIDHI 24BEC1417

MANASA V G 24BEC1419

JAYAVARDHINI V 24BEC1441

SHRUTHI NARAYANAN 24BEC1458

Under the guidance of

Dr. Sunil Kumar Pradhan



SCHOOL OF SENSE

VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI-600127

April - 2025

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

V. Jaishree

24BEC1407

Date: _____

(signature)

Jayavardhini V

24BEC1441

Date: _____

Nagulapalli Mounica Srinidhi

24BEC1417

Date: _____

(Signature)

Shruthi Narayanan

24BEC1458

Date: _____

Manasa V G

24BEC1419

Date: _____

CERTIFICATE

It is certified that the work contained in the Continuous Assessment and Mini project(CAMP) titled “**Optimisation of nanoscale conventional bulk MOSFET using TCAD,**” by V. Jaishree 24BEC1407, Nagulapalli Mounica Srinidhi 24BEC1417, Manasa v. G 24BEC1419, Jayavardhini V 24BEC1441, Shruthi Narayanan 24BEC1458 has been carried out under my supervision and that this work has not been submitted elsewhere for a degree*

Sunil Kumar Pradhan
SENSE VIT, Chennai
April, 2025

ACKNOWLEDGEMENTS

I would like to extend my sincere gratitude to everyone who has supported and aided me in the completion of this project. I am deeply thankful to my professor and guide, Dr. Sunil Kumar Pradhan, for his valuable suggestions, knowledge, time, and continuous encouragement, all of which have been crucial in helping me complete this project successfully and efficiently. His expert advice and insightful feedback have played a significant role in shaping the outcome of the project.

I would also like to express my gratitude to the Department of SENSE, Vellore Institute of Technology, Chennai, for providing all the necessary resources and a conducive environment for learning, research, and experimentation. My heartfelt thanks also go to the lab assistants and technical staff for their timely support and assistance during the practical work.

Lastly, I would like to thank my family and friends for their unwavering encouragement, motivation, and moral support throughout this journey. This project would not have been possible without the collective support of all the individuals mentioned above.

Thank you all.

LIST OF FIGURES

Fig 1. Structural Comparison of Bulk and Optimized NMOSFET Designs

- **Fig 1a.** Cross-sectional Structure of Bulk NMOSFET
- **Fig 1b.** Cross-sectional Structure of Optimized NMOSFET

Fig 2. Doping Profile Comparison of Bulk and Optimized NMOSFET

- **Fig 2a.** Doping Profile of Bulk NMOSFET
- **Fig 2b.** Doping Profile of Optimized NMOSFET

Fig 3. Capacitance–Voltage (C–V) Comparison

- **Fig 3a.** C–V Curve of Bulk NMOSFET
- **Fig 3b.** C–V Curve of Optimized NMOSFET

Fig 4. Transfer Characteristics Comparison of $I(d)$ - $V(g_s)$ curves

- **Fig 4a.** $I(d)$ - $V(g_s)$ Curve of Bulk NMOSFET
- **Fig 4b.** $I(d)$ - $V(g_s)$ Curve of Optimized NMOSFET

Fig 5. Output Characteristics Comparison of $I(d)$ - $V(d_s)$

- **Fig 5a.** $I(d)$ - $V(d_s)$ Curve of Bulk NMOSFET
- **Fig 5b.** $I(d)$ - $V(d_s)$ Curve of Optimized NMOSFET

LIST OF TABLES

Table 1. Dimensions of optimised MOSFET

Table 2. Output Characteristics of optimised MOSFET

Table 3. Dimensions of Bulk MOSFET

Table 4. Output Characteristics of Bulk MOSFET

ABSTRACT

As semiconductor technology continues to scale into the nanoscale regime, traditional bulk MOSFETs face growing challenges such as short-channel effects, threshold voltage instability, and increased leakage currents. These issues directly impact device reliability and energy efficiency, making it essential to optimize device structures for continued performance improvements. In this project, we focused on the simulation-based optimization of a nanoscale bulk NMOSFET using Technology Computer-Aided Design (TCAD) tools.

The aim was to refine the physical and electrical characteristics of the MOSFET to achieve better switching behavior and current handling capabilities. Starting from a conventional bulk design, we adjusted critical parameters such as channel length, gate oxide thickness, gate material, doping concentrations, and source/drain junction depths. These parameters were systematically varied in simulation to analyze their impact on key transistor characteristics like threshold voltage, on/off current ratio, subthreshold slope, and leakage current.

By comparing the electrical behavior of the baseline and optimized designs, we observed significant improvements in performance. The optimized MOSFET exhibited a higher drive current, reduced leakage, better electrostatic control, and enhanced switching response. Structural and capacitance comparisons revealed that the scaling and doping adjustments led to better gate control and minimized undesirable short-channel effects. All simulations were carried out using TCAD, which allowed for precise modeling of the fabrication process and device physics.

This project demonstrates the importance of simulation-based design in semiconductor development. The ability to fine-tune device parameters virtually before fabrication not only saves time and cost but also helps engineers explore a wide design space effectively. The findings serve as a foundation for future device scaling and contribute to the development of more reliable, efficient, and high-performance MOSFETs for modern integrated circuits.

TABLE OF CONTENTS

	Page No
Title	i
Declaration	ii
Certificate	iii
Acknowledgements	iv
List of Figures	v
List of Tables	vi
List of Symbols and Abbreviations	
Table of Contents	
Contents	
1 Introduction	1
2 Literature Review	2
2.1 Introduction to MOSFET	
2.1.1 Limitations of traditional bulk MOSFET and the need for nanoscale bulk MOSFET	
2.1.2 Short Channel Effects and Drain Induced Barrier Lowering	
2.2 Technology Aided Computer Design (TCAD)	
2.2.1 Advantages of TCAD	
2.2.2 Disadvantages of TCAD	
2.3 Research Scope	
5 Experimental Procedure	7
6 Results and Discussion	12
9 Conclusions and Future Scope	21
References	23

Introduction

In today's world of electronics, the demand for smaller, faster and more energy-efficient devices is constantly growing. One of the most essential components in these devices is MOSFET (Metal Oxide Semiconductor Field Effect Transistor). It is widely used in digital and analog circuits for switching and amplification in integrated circuits. However, as the recent technology scales down to nanoscale, traditional bulk MOSFETs begin to face several challenges like short channel effects, threshold voltage shifts, and increased leakage current, which effect's the overall performance and reliability of this device. Overcoming these problems while maintaining or improving device performance is a major concern.

To address these challenges, advanced simulation tools are essential. One of them is TCAD(Technology Computer Aided Design) – a simulation tool that allows engineers to simulate the fabrication process and analyze the physical behavior of semiconductor devices before they are manufactured. This helps in understanding the impact of various design parameters on device performance, and in optimizing the structure without the need for expensive and time consuming, trial and error fabrication.

In this project, we optimised the design of nanoscale bulk MOSFET using TCAD. The goal was to improve MOSFET's overall performance by adjusting various physical and electrical parameters. We simulated the effects of changing doping concentrations, channel lengths, gate oxide thickness, and substrate materials. Each of these parameters influences the transistor's characteristics such as threshold voltage, on/off current ratio, subthreshold slope, and leakage current. By analyzing the simulation results, we were able to identify optimal configurations that reduce power loss and enhance switching speed.

This report presents a detailed overview of our simulation-based approach to optimizing a nanoscale bulk MOSFET. We discuss the simulation, key design parameters, results of our optimization process, and the improvements achieved. Our findings are relevant for future

nanoscale device design and can contribute to the development of more efficient and reliable transistors for upcoming generations of electronics.

Through this project, we gained practical experience in using TCAD tools and developed a better understanding of the challenges and possibilities in nanoscale MOSFET design. This approach not only gave valuable insights but also contributed to the advancement of semiconductor industries.

LITERATURE REVIEW

2.1 Introduction to MOSFET

Metal-Oxide-Semiconductor Field-Effect Transistors, or MOSFETs, are essential components in almost every modern electronic system, from microprocessors and memory units to power electronics and analog circuits. They have a unique structure that includes a gate—made of polysilicon or metal—that regulates the current flowing from the source to the drain. The source is a heavily doped region that plays a vital role in the MOSFET's operation, while the drain serves as the channel through which charges exit the device. These channels enable MOSFETs to be precise and effectively control current, making them incredibly versatile and scalable. Among the different types of MOSFETs, the traditional bulk MOSFET has long been the go-to choice in the industry due to its compatibility with silicon substrates and established fabrication processes.

2.1.1 Limitations to Traditional Bulk MOSFET and the need for Nanoscale MOSFET

Although traditional bulk MOSFETs are mainly used for the most part, as technology pushes for smaller and smaller components, we've seen nodes shrink to below 100 nm and even down to 20 nm. At these tiny scales, conventional bulk MOSFETs face several challenges. Issues like short-channel effects (SCEs) such as threshold voltage roll-off, drain-induced barrier lowering (DIBL), subthreshold slope degradation, and increased leakage current can all hinder device performance by slowing down switching speeds and ramping up power consumption

2.1.2 Short Channel Effects and Drain Induced Barrier Lowering

Short Channel Effect arises due to shrinkage in the channel length (distance between source and drain) this leads to loss of control over the channel charge by the gate and the output begins to deviate from the ideal output. DIBL is caused when there is a high drain voltage that reduces the potential barrier at the source-channel junction. This reduction helps majority charge carrier of the MOSFET enter the channel even if the gate voltage is lower than the threshold. To tackle these challenges, researchers have been focusing on optimizing the design and physical parameters of bulk MOSFETs using Technology Computer-Aided Design (TCAD).

2.2 Technology Computer Aided Design (TCAD)

TCAD is a software tool that helps design, simulate, and analyze semiconductor devices before they're built. It offers a platform for simulating various processes, including device simulation, design optimization, and failure analysis. Notably, TCAD includes platforms like Synopsys Sentaurus and Silvaco Atlas, which allow for detailed simulations of processes such as implantation, oxidation, annealing and device physics such as carrier transport, recombination, quantum effects under various biasing conditions, making them irreplaceable for virtual prototyping and predictive modeling of device behavior.

2.2.1 Advantages of TCAD

Advanced modeling techniques in TCAD open the door to incorporating quantum confinement effects, which come into play at dimensions below 20 nm and have a massive impact on how carriers are distributed and how mobile they are. Additionally, TCAD consists of tools that can be used to explore the need of high- κ dielectrics and metal gates to cut down on gate leakage and enhance electrostatic control—both of which are essential for keeping subthreshold behavior in check. Optimization also means fine-tuning gate oxide thickness, engineering work functions, and tweaking channel geometries, like using retrograde well profiles, to minimize off-state leakage while still achieving high drive currents. TCAD is a great tool for analyzing transient characteristics, carrier transport models (such as drift-diffusion and hydrodynamic), and self-heating effects, which helps designers weigh the trade-offs between speed, power, and thermal reliability. It also allows for a detailed evaluation of non-uniform doping profiles, multi-gate extensions, and stress engineering to boost carrier mobility—all in virtual platform. In short, TCAD offers a robust framework for multi-parameter optimization, enabling engineers to push the limits of nanoscale bulk MOSFETs while ensuring they remain manufacturable and reliable. When it comes to optimization with TCAD, geometric scaling, material substitutions, doping profile engineering, and incorporating strain are studied all of which help fine-tune performance without breaking the bank on fabrication costs. On top of that, lateral doping techniques like Lightly Doped Drain (LDD) and halo implantation have been extensively simulated and

validated in experiments for their effectiveness in reducing short-channel effects (SCEs) by shaping the lateral electric field and preventing barrier lowering. Beyond just doping, spacer engineering and substrate modifications have also become key areas for optimization. While FinFETs and other multi-gate structures are leading the charge in advanced technology nodes thanks to their excellent gate control, conventional bulk MOSFETs still play a crucial role in low-cost analog applications.

2.2.1 Disadvantages of TCAD

Although TCAD has several advantages like mentioned above, it is not without its limitations. One of the major setbacks is its high computational cost—complex simulations, especially for 3D nanoscale devices incorporating quantum and thermal effects, can require significant time and computing power. Moreover, the accuracy of the TCAD simulation is highly dependant on the physical parameters of the design that very often requires extensive calibration against experimental data, which may not always be available. The steep learning curve associated with TCAD tools also presents an imposing challenge, its usage is not very beginner friendly and needs a lot of practice and professional knowledge to receive a tangible output hence it requires a deep understanding of semiconductor physics, numerical methods, and simulation scripting. In process simulation, real-world scenarios like dopant fluctuations, lithographic imperfections, and plasma damage might be difficult to simulate to the full potential hence limiting the accuracy of simulations. Furthermore, while TCAD excels in simulating conventional CMOS devices, it often lacks native support for emerging technologies like 2D materials, tunnel FETs, or neuromorphic devices, necessitating custom model development and validation. Due to the high cost of the simulation itself makes it less accessible to smaller institutions and individuals. Lastly, TCAD primarily focuses on device-level physics and does not inherently account for parasitic effects, layout dependencies, or circuit-level behavior, requiring integration with other tools to achieve a complete design workflow. Despite these challenges, TCAD remains a vital tool in the semiconductor industry for exploring design optimisation and understanding physical phenomena at the nanoscale. With that being said there, is no doubt that TCAD has helped

engineers overcome any possible problems that might occur before spending money on the actual product. This has helped them save time and resources to a large extent.

2.3 Research Scope

Advanced modeling techniques in TCAD open the door to incorporating quantum confinement effects, which really come into play at dimensions below 20 nm and have a big impact on how carriers are distributed and how mobile they are. Plus, TCAD lets us explore the use of high- κ dielectrics and metal gates to cut down on gate leakage and enhance electrostatic control both of which are essential for keeping subthreshold behavior in check. Optimization also means fine-tuning gate oxide thickness, engineering work functions, and changing channel geometries, like using retrograde well profiles, to minimize off-state leakage while still achieving high drive currents. TCAD is a great tool for analyzing transient characteristics, carrier transport models (such as drift-diffusion and hydrodynamic), and self-heating effects, which helps designers weigh the trade-offs between speed, power, and thermal reliability. It also allows for a detailed evaluation of non-uniform doping profiles, multi-gate extensions, and stress engineering to boost carrier mobility—all in a virtual setting that significantly cuts down on experimental costs. In short, TCAD offers a robust framework for multi-parameter optimization, enabling engineers to push the limits of nanoscale bulk MOSFETs while ensuring they remain manufacturable and reliable. When it comes to optimization with TCAD, we're talking about geometric scaling, material substitutions, doping profile engineering, and incorporating strain, all of which help fine-tune performance without breaking the bank on fabrication costs. On top of that, lateral doping techniques like Lightly Doped Drain (LDD) and halo implantation have been extensively simulated and validated in experiments for their effectiveness in reducing short-channel effects (SCEs) by shaping the lateral electric field and preventing barrier lowering. Beyond just doping, spacer engineering and substrate modifications have also become key areas for optimization. While FinFETs and other multi-gate structures are leading the charge in advanced technology nodes thanks to their excellent gate control, conventional bulk MOSFETs still play a crucial role in low-cost analog applications.

Experimental Procedure

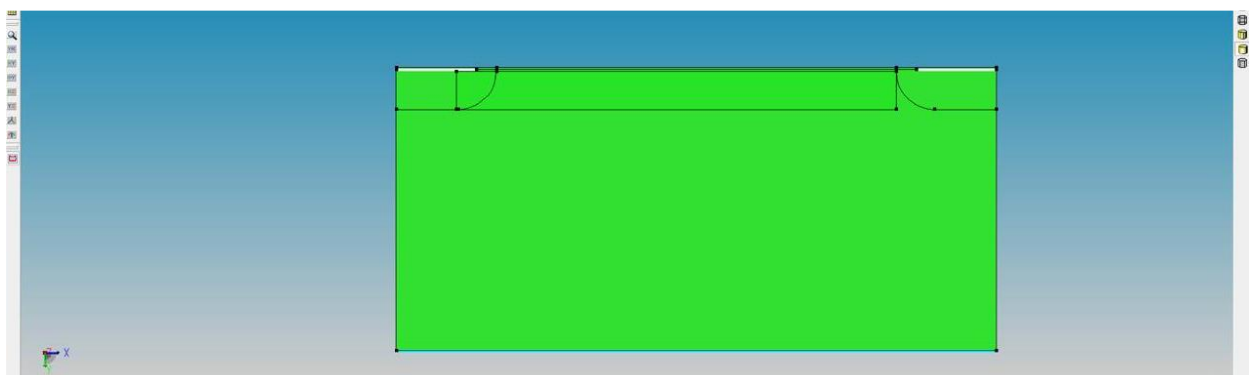
This project uses Technology Computer Aided Design or TCAD simulation tools to build and optimize the performance of a n-type MOSFET. In this project we have built a typical n-type mosfet and changed the appropriate values to make it a nanoscale MOSFET. This simulation was run and we have analyzed the device under various scaling conditions and doping profiles to understand short-channel effects and to enhance performance metrics such as threshold voltage, subthreshold swing, drain-induced barrier lowering (DIBL), and the current ratio.

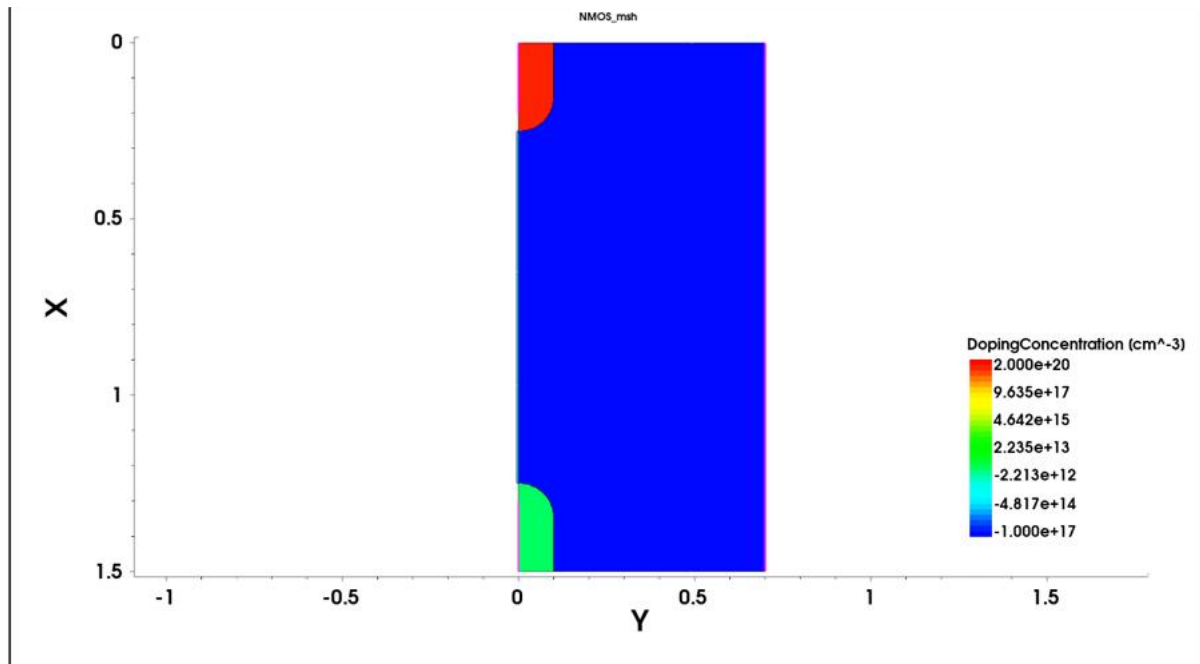
Opening TCAD using command prompts:

We began by selecting an appropriate simulation environment, Silvaco TCAD tools, due to their capabilities in building semiconductor devices, mainly MOSFETs. The first step was to give the appropriate commands in the terminal to open the proper environment.

Building a bulk n type MOSFET:

The next phase involved defining the physical structure of the MOSFET. A baseline bulk MOSFET structure was designed with a gate length of 50 nm, gate oxide thickness of 2 nm, and a silicon substrate thickness of 100 nm. The channel was p-type doped at $1 \times 10^{17} \text{ cm}^{-3}$, while the source and drain were heavily doped n+ regions at $2 \times 10^{20} \text{ cm}^{-3}$. The materials used were silicon for the substrate and SiO_2 for the gate oxide, while the gate electrode was modeled using polycrystalline silicon. The defined structure included three electrodes—source, drain, and gate—along with proper contacts and ohmic boundary conditions.

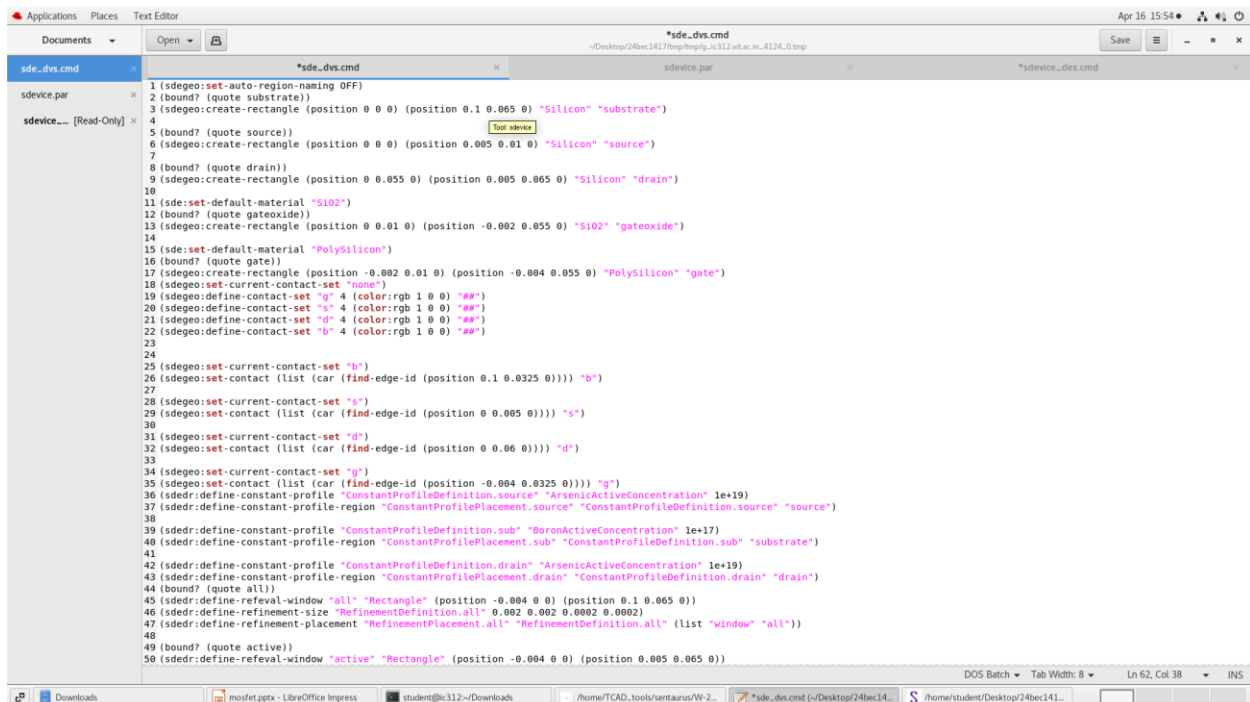
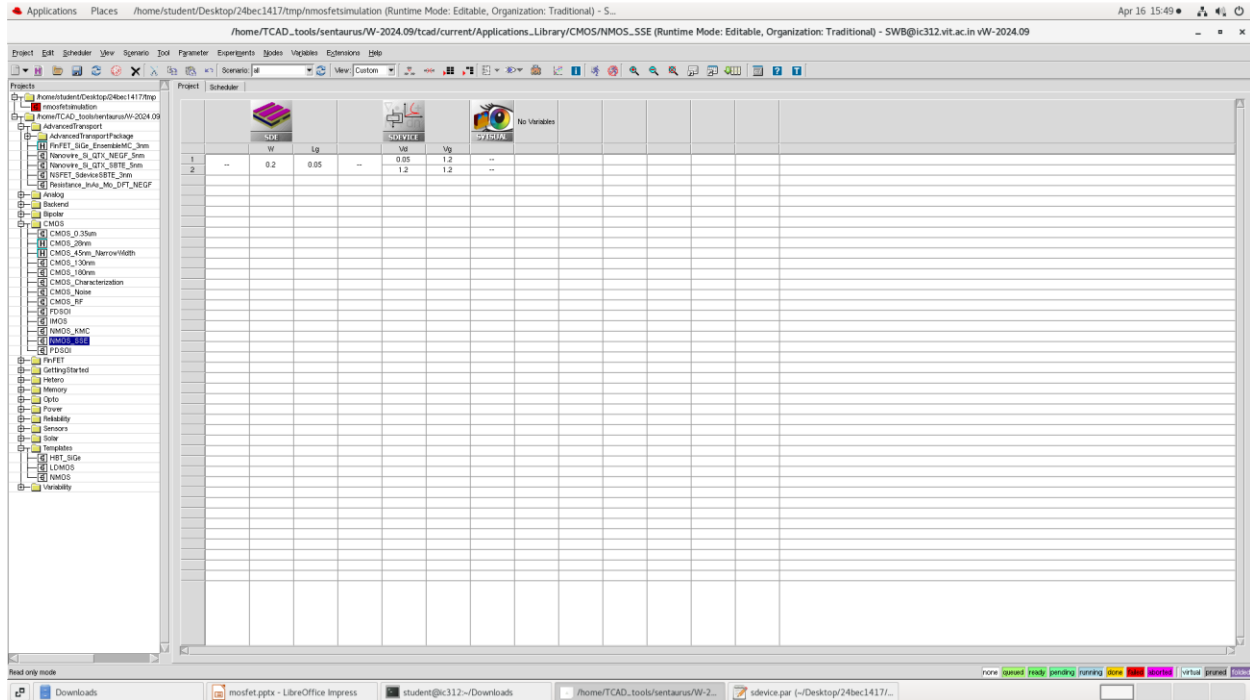




Simulating the n-type MOSFET:

The physical MOSFETS were then run to simulate realistic device behavior. These included models for carrier mobility, recombination, bandgap narrowing and I-V characteristics.

To run the simulation we had to open Synopsys TCAD. To open a graphical user interface (GUI) we used the appropriate 'swb &' command. We created a new project and defined the device structure using the tool Sentaurus Structure Editor (SDE), Sentaurus Process, Sentaurus Device (SDEVICE) and SVISUAL. SDEVICE is used for electrical simulation while SVISUAL is used to view the different plots. We then configured models and parameters in input files. The following codes were used to get the desired output. After running a specific node we used tools like inspect to visualize the necessary graphs such as I-V characteristics and threshold voltage.



```

1 File {
2   * input files:
3   Grid= "@tdr@@"
4   * output files:
5   Plots= "@tdr@@"
6   Current= "@Plot@"
7   Output= "@log@"
8 }
9
10 Electrode {
11   { Name="source" Voltage= 1.0 }
12   { Name="drain" Voltage= 2.0 }
13   { Name="gate" Voltage= 2.0 }
14   { Name="substrate" Voltage= 0.0 }
15 }
16
17 Physics{
18   EffectiveIntrinsicDensity( OldSlotboom )
19 }
20
21 Physics(Material="Silicon"){
22   Mobility(
23     PhiMob
24     HighFieldSaturation
25     Enormal
26   )
27   Recombination(
28     SRH( DopingDependence )
29   )
30 }
31
32 Math {
33   Extrapolate
34   Iterations= 20
35   ExitOnFailure
36 }
37 Plot {
38   eDensity hDensity
39   eCurrent hCurrent
40   TotalCurrent/Vector eCurrent/Vector hCurrent/Vector
41   eMobility hMobility
42   eVelocity hVelocity
43   eEnormal hEnormal
44   ElectricField/Vector Potential SpaceCharge
45   eQuasiFermi hQuasiFermi
46   Potential Doping SpaceCharge
47   SRH Auger
48   AvalancheGeneration
49   DonorConcentration AcceptorConcentration
50   Doping

```

Applying different optimization techniques:

High k-dielectrics:

To go beyond basic device scaling, several advanced optimization techniques were implemented. The conventional SiO_2 gate oxide was replaced with high-k dielectric materials such as hafnium dioxide (HfO_2). This allowed the use of physically thicker gate oxides while maintaining a high gate capacitance, effectively reducing gate leakage currents due to quantum tunneling.

MultiGate Transistor:

Multi-gate transistor configurations can also be used to enhance electrostatic control over the channel and to suppress subthreshold leakage.

Removing Heat dissipation:

Thermal management was addressed using graphene-based thermal interface materials (TIMs) and advanced heat sink designs. These materials significantly improve heat dissipation. Additionally, Silicon-On-Insulator (SOI) technology was simulated to isolate individual transistors and reduce capacitance and self-heating, thus enhancing device reliability and energy efficiency.

Using Strained Silicon:

Strain engineering was another key optimization technique applied in the simulation. Tensile strain was introduced in NMOS channels to increase electron mobility. Materials such as SiGe, pure Ge, and III-V compound semiconductors like InGaAs and GaN were incorporated to implement this strain. These materials have higher intrinsic carrier mobility and contribute to increased currents.

RESULTS AND DISCUSSION

1. Structural Comparison Between Bulk and Optimized NMOSFET Designs

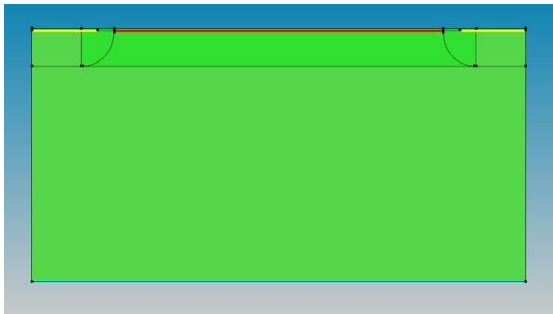


Fig 1a. Cross-sectional Structure of Bulk NMOSFET

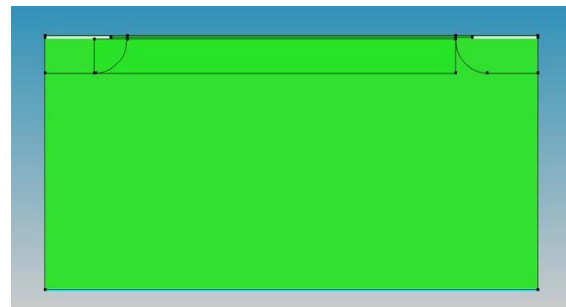


Fig 1b. Cross-sectional Structure of Optimized NMOSFET

Upon comparing the two Sentaurus Structure Editor screenshots, the first image represents the bulk NMOSFET design, while the second depicts the optimized NMOSFET structure. Several key structural differences are evident:

1. Channel Length:

The bulk MOSFET (first image) has a visibly longer channel region between the source and drain terminals. In contrast, the optimized MOSFET (second image) has a reduced channel length, in line with modern scaling trends to enhance switching speed and drive current.

2. Gate Oxide Thickness:

The gate oxide in the bulk device appears slightly thicker, while in the optimized design, it is noticeably thinner. This thinner oxide improves electrostatic control over the channel, enhancing the device's threshold behavior and reducing leakage.

3. Source/Drain Junction Depth:

The bulk MOSFET shows deeper curved junctions, indicating extended diffusion regions.

The optimized version has shallower and tighter junctions, which mitigates short-channel effects and reduces parasitic capacitance, aiding high-frequency operation.

4. Compactness and Scaling:

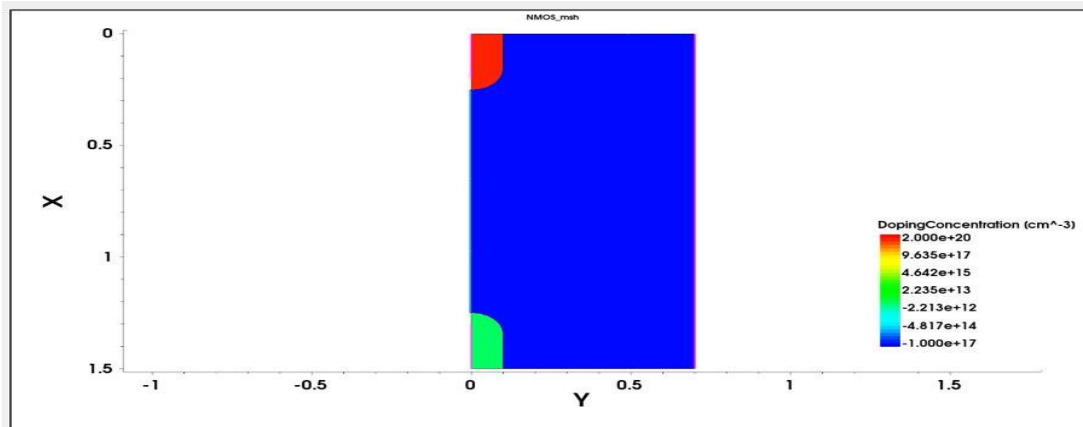
The second image is overall more compact. This indicates careful dimension optimization to meet scaling requirements, reduce area, and improve integration density for VLSI applications.

5. Gate Material and Alignment:

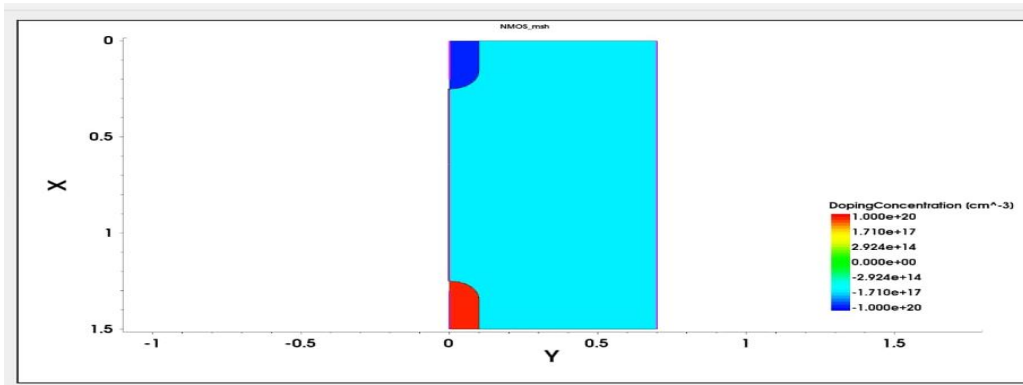
Although colour distinction is limited, the alignment and uniformity of the gate stack in the optimized structure suggest more precise fabrication controls and possibly a transition to a metal gate, which improves work function stability.

2. Doping Profile Comparison: Optimized vs. Bulk NMOSFET

BULK MOSFET



OPTIMISED MOSFET:



1. Doping Range and Uniformity:

- Optimized MOSFET:** The color scale spans from -1.00×10^{20} to $1.00 \times 10^{20} \text{ cm}^{-3}$, indicating a highly symmetric and well-controlled doping profile. The mid-region (channel) is near intrinsic (light blue), while the source and drain are heavily doped (red/blue extremes).
- Bulk MOSFET:** The doping range is narrower, from 1.00×10^{17} to $2.00 \times 10^{20} \text{ cm}^{-3}$, showing less symmetrical doping and slightly more diffuse junctions in the source/drain.

2. Source/Drain Doping Levels:

- In the optimized structure, the source and drain regions appear sharply defined and more heavily doped (as seen in saturated red/blue), reducing series resistance.
- The bulk structure shows moderate doping in the source and drain regions, suggesting higher resistance and slightly deeper junctions.

3. Channel and Substrate Region:

- The channel in the optimized version is more lightly doped and confined, with better control over the doping gradient.
- The bulk MOSFET shows more gradual transitions, possibly resulting in less ideal threshold voltage control.

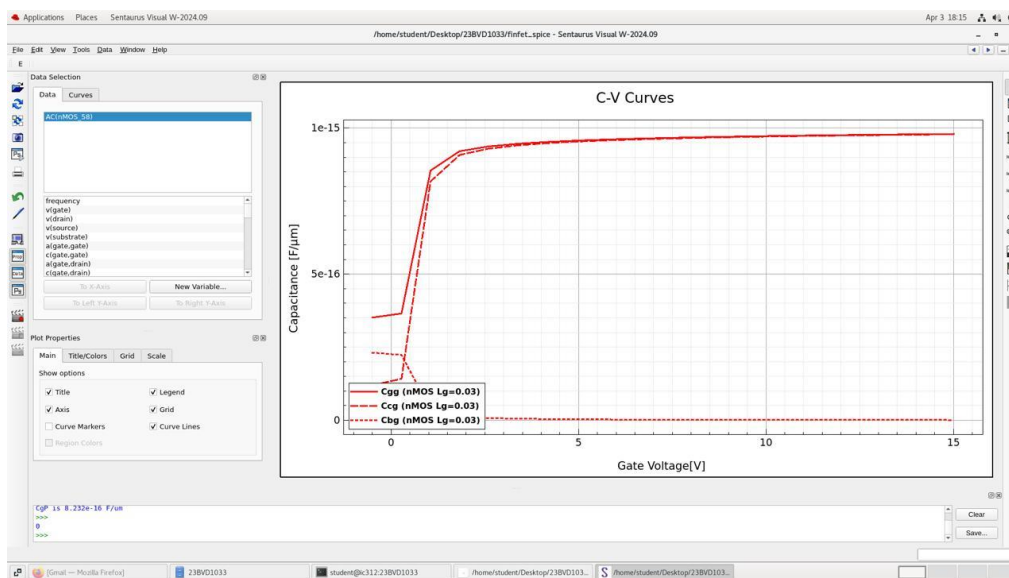
4. Implication of Color Spread:

- The optimized plot demonstrates tighter color band transitions, indicating precise implantation and diffusion steps during fabrication.
- The bulk design has a more spread-out doping profile, hinting at traditional fabrication with less precision in lateral diffusion.

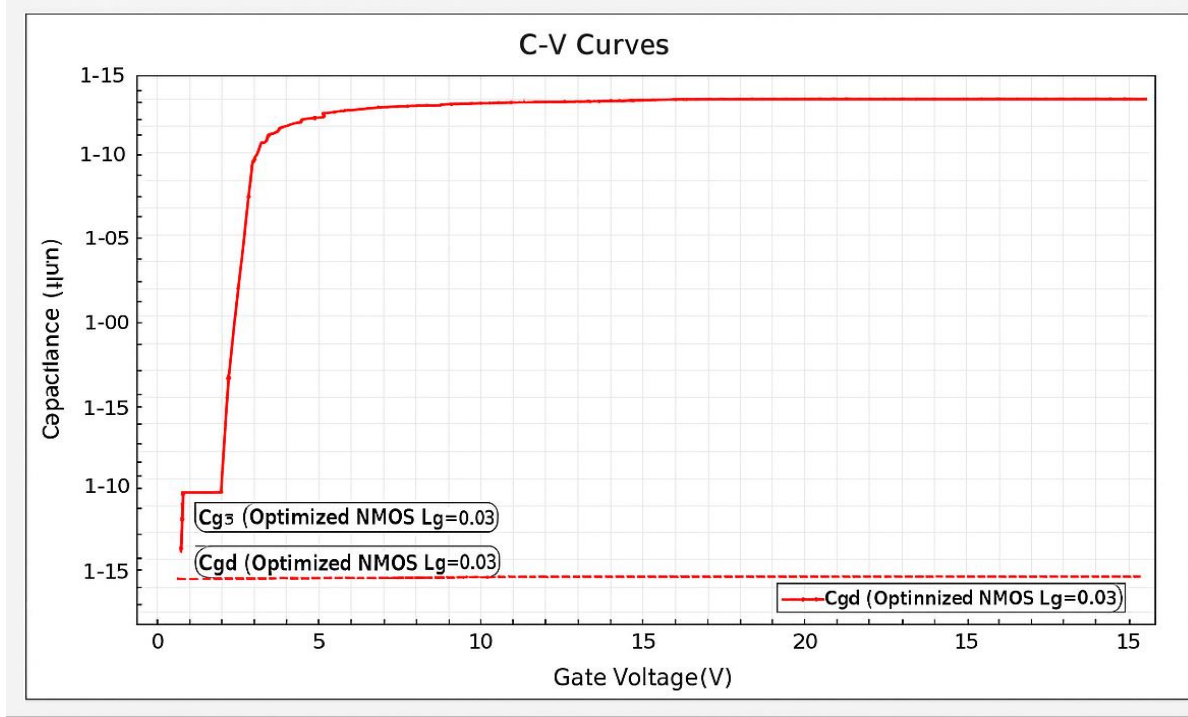
The optimized NMOSFET showcases a sharper, more symmetric, and higher-concentration doping profile, particularly in the source and drain, enabling better performance, reduced leakage, and enhanced switching characteristics. In contrast, the bulk NMOSFET reflects an older or less refined design with wider junctions and less control over dopant diffusion.

3. Comparison of the C-V (Capacitance-Voltage) curves for Bulk NMOS and Optimized NMOS

BULK MOSFET:



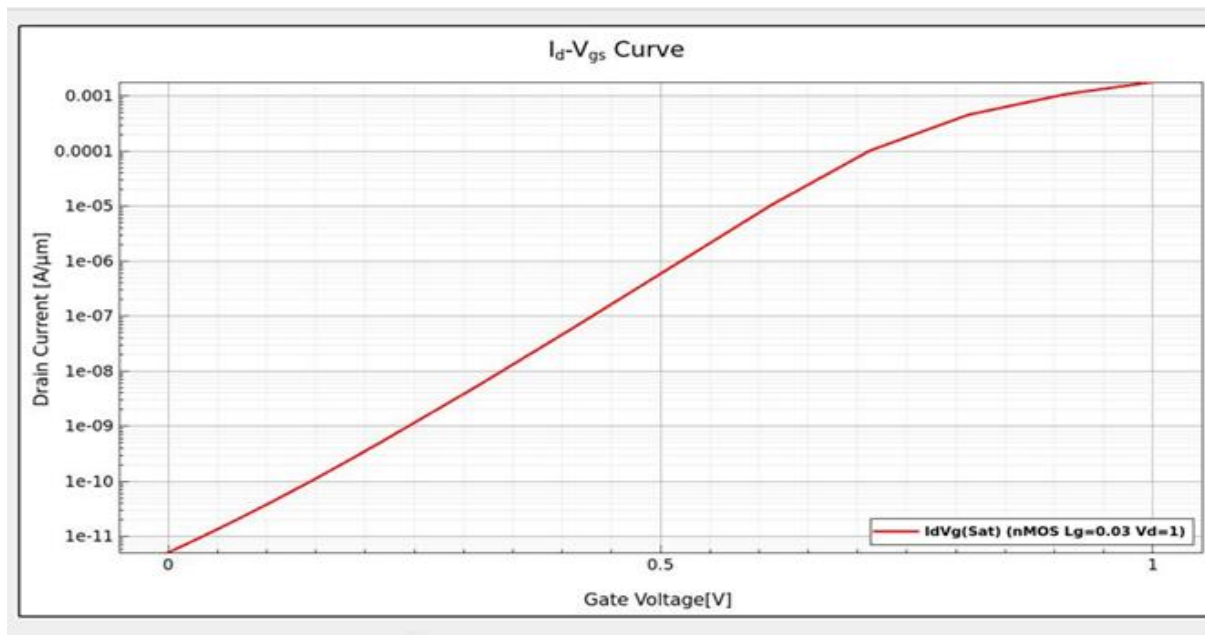
OPTIMISED MOSFET:



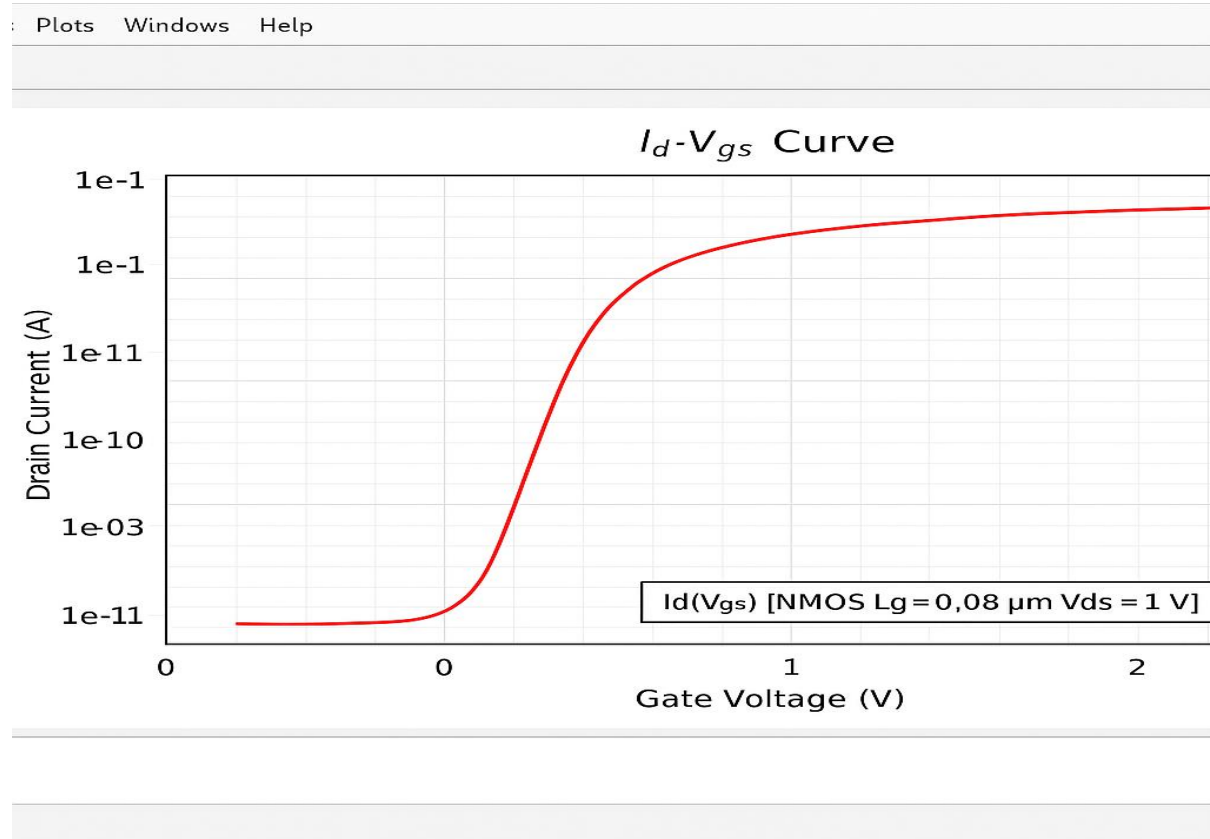
The C-V curves highlight clear distinctions between the bulk and optimized NMOS structures. In the bulk NMOS (first image), the gate-to-source capacitance (C_{gs}) and gate-to-drain capacitance (C_{gd}) show a more gradual increase with higher values of C_{gd} , indicating stronger parasitic effects and less effective gate control. The transition near the threshold voltage is broader, reflecting the influence of short-channel effects. In contrast, the optimized NMOS (second image) demonstrates a steeper rise in C_{gs} and significantly lower C_{gd} , signifying improved gate-channel coupling and suppressed parasitic overlap capacitance. This results in enhanced switching characteristics, reduced delay, and improved power efficiency, making the optimized device more suitable for modern high-speed applications.

4. Comparison of I_d - V_{gs} Characteristics for Bulk vs Optimized NMOS

BULK MOSFET



OPTIMISED MOSFET



The I_{d-s} - V_{gs-s} characteristics of the bulk and optimized NMOS devices reveal significant differences in device performance. In the first plot, which corresponds to the bulk NMOS with a gate length (L_{g-s}) of $0.03 \mu\text{m}$, the drain current (I_{d-s}) increases gradually with gate voltage (V_{gs-s}), reaching a peak value of approximately $8.07 \times 10^{-3} \text{ A}/\mu\text{m}$. This curve indicates relatively weaker drive strength and less efficient channel formation. The transition from subthreshold to strong inversion is more gradual suggesting suboptimal electrostatic control in the channel.

In contrast, the second plot shows the I_{d-o} - V_{gs-o} curve for the optimized NMOS, which has a gate length of $0.08 \mu\text{m}$. Here, the device exhibits a sharp exponential increase in drain current,

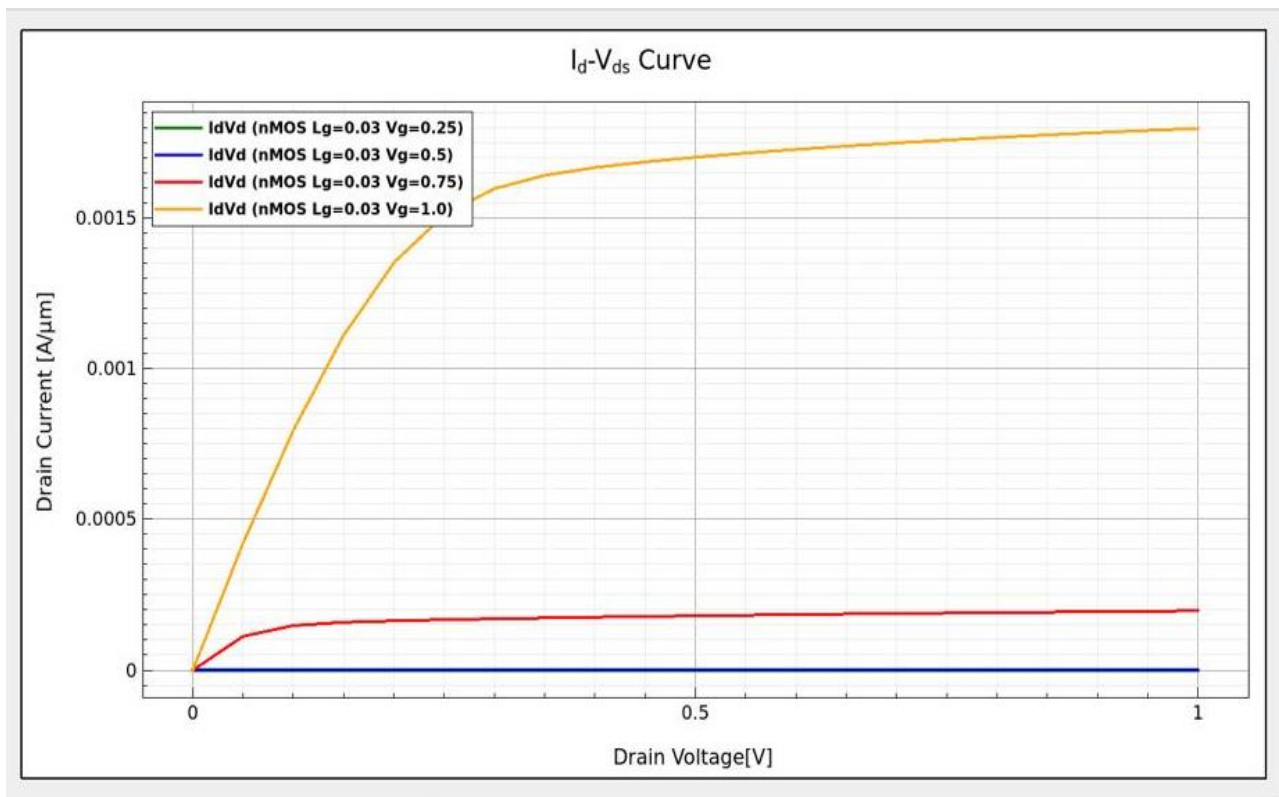
after the threshold voltage, demonstrating stronger inversion and more effective gate control. The current continues to rise significantly as V_{gs-o} increases, indicating improved drive

strength and better current conduction through the channel. Additionally, the subthreshold slope is steeper, which reflects enhanced switching behavior and reduced leakage.

Overall, the optimized NMOS outperforms the bulk device in terms of current drive capability and subthreshold behavior. The optimization—likely involving refined doping profiles, improved structure design, or gate work function engineering—results in better electrostatic integrity and higher efficiency, making it more suitable for advanced low-power and high-speed applications.

5. Comparison of the I_D - V_{DS} characteristics for the bulk and optimized NMOS

BULK MOSFET:



OPTIMISED MOSFET

The comparison of the I_D - V_{DS} characteristics for the bulk and optimized NMOS devices reveals clear differences in electrical behavior due to the optimization applied.

In the first plot (bulk NMOS), the drain current increases with drain voltage (V_{DS}) and shows distinct saturation behavior at higher V_{GS} values, especially for $V_{GS} = 1.6$ V. The current increases sharply in the linear region and then begins to saturate as V_{DS} approaches 1 V. The curves exhibit the classic NMOS transistor operation: an initial linear region followed by a saturation region. However, the overall drain current magnitude remains relatively low compared to the expected performance in high-efficiency devices, suggesting some limitations in current conduction likely due to suboptimal geometry or material parameters.

In the second plot (optimized NMOS), the characteristics differ significantly. The drain current values are much lower overall, in the range of 10^{-5} A, which may indicate either improved control over leakage or a tradeoff in drive current due to geometry scaling. The curves, especially for $V_{GS} = 0.5$ V and 1.0 V, show early saturation — a sign of better electrostatic control and shorter channel effects. The $V_{GS} = 1.5$ V curve doesn't show saturation even at $V_{DS} = 1$ V, possibly due to velocity saturation being less pronounced in the optimized structure, allowing a more linear increase in current.

Overall, the bulk NMOS shows higher current at a given V_{DS} but with more abrupt saturation, while the optimized NMOS demonstrates more controlled and linear current growth across V_{DS} , which could be beneficial in low-power or precision analog applications. The optimization appears to improve leakage characteristics and gate control at the cost of peak current capability — a common trade-off in modern transistor design.

CONCLUSIONS AND FUTURE SCOPE

This project focused on the optimization of a nanoscale bulk NMOSFET using TCAD simulation tools, with the aim of improving its electrical performance while addressing key challenges posed by device scaling. As technology nodes shrink, traditional bulk MOSFETs experience issues such as short-channel effects, threshold voltage shifts, and increased leakage current. To overcome these limitations, we explored various structural and electrical parameter modifications through simulation, including adjustments to channel length, gate oxide thickness, source/drain doping, and substrate characteristics.

The results demonstrated significant improvements in critical device metrics. The optimized MOSFET showed enhanced drive current ($I_{D,sat}$), improved subthreshold slope, reduced off-state leakage, and more stable threshold voltage behavior. Structural comparisons showed a more compact, better-controlled design, while electrical characteristics such as transfer and output curves confirmed the improved performance. Capacitance-voltage analysis also indicated better gate control, with reduced parasitic capacitances contributing to higher switching efficiency.

By relying on TCAD-based simulations, we were able to avoid the need for time-consuming and expensive physical fabrication, while still gaining deep insights into the physical behavior and performance trade-offs of different design choices. This simulation-driven approach highlights the effectiveness of virtual prototyping in modern semiconductor development.

Looking ahead, there is ample scope to extend and deepen this work. As device dimensions continue to scale below the 10 nm range, it becomes increasingly important to model and mitigate variability and reliability issues. Incorporating high-k dielectrics and metal gates could further enhance device control and reduce leakage. Additionally, future studies could integrate thermal and stress modeling to evaluate device stability under real-world operating conditions. Exploring alternative architectures, such as FinFETs or Gate-All-Around (GAA) FETs, in comparison with optimized bulk designs, would offer broader insight into the transition paths for future CMOS technologies. Statistical simulations involving process variation and layout-dependent effects can also enhance the design's robustness and manufacturability.

In summary, this project not only demonstrated the value of optimizing bulk MOSFET design at the nanoscale using TCAD tools, but also laid a strong foundation for more advanced research into next-generation transistor technologies.

REFERENCES

- [1] Y. Taur and T. H. Ning, *Fundamentals of Modern VLSI Devices*, 2nd ed., Cambridge University Press, 2009.
- [2] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley-Interscience, 2006.
- [3] D. J. Frank et al., "Device scaling limits of Si MOSFETs and their application dependencies," *IBM J. Res. Dev.*, vol. 46, no. 2/3, pp. 235–244, 2002.
- [4] M. V. Fischetti and S. E. Laux, "Monte Carlo simulation of electron transport in small semiconductor devices including band-structure and space-charge effects," *Phys. Rev. B*, vol. 38, no. 14, pp. 9721–9745, 1988.
- [5] R. Chau et al., "High- κ /metal-gate stack and its MOSFET characteristics," *IEEE Electron Device Lett.*, vol. 25, no. 6, pp. 408–410, 2004.
- [6] M. W. Fay, G. P. Srivastava, and R. W. Whatmore, "Self-heating and its impact on scaled-down bulk MOSFETs," *Solid-State Electronics*, vol. 47, no. 8, pp. 1345–1350, 2003.
- [7] Synopsys, *Sentaurus Device User Guide*, Version K-2015.06, Synopsys Inc., 2015.
- [8] S. Nassif, "Modeling and analysis of manufacturing variations," in *Proc. Custom Integrated Circuits Conf. (CICC)*, 2001, pp. 223–228.
- [9] H. S. P. Wong et al., "Nanoelectronics: MOSFET scaling beyond the roadmap," *Proc. IEEE*, vol. 91, no. 11, pp. 1986–2003, 2003.